



## CALIFORNIA STATE UNIVERSITY, LONG BEACH

DEPARTMENT OF CIVIL ENGINEERING & CONSTRUCTION ENGINEERING MANAGEMENT

December 1, 2025

United States Citizenship and Immigration Services

**Subject:** Letter of Support for Mr. Vivek Mishra's EB-1A Petition

To whom it may concern,

My name is Dr. Shailesh Chandra. I serve as Director of the Mobility Research and Innovation Lab, Director of Education and Workforce Development for the Center for Freight Transportation for Efficient & Resilient Supply Chain (FERSC) at the University of Tennessee, Knoxville, and Associate Professor of Civil and Transportation Engineering at California State University, Long Beach. For more than fifteen years, I have directed state- and federally funded research in transportation planning, infrastructure systems, and emerging mobility technologies.

It is my privilege to recommend Mr. Vivek Mishra, whose contributions to our research program and the broader field have been both distinctive and far-reaching. I first met Vivek in January 2020 when he joined my lab as a research assistant on a project funded by the California Department of Transportation (Caltrans). The project he supported was financed through Caltrans and complemented by grants from the U.S. Department of Transportation (USDOT), the Department of Homeland Security, and the California State University Office of the Chancellor. These funding streams, documented in Mineta Transportation Institute technical reports and associated publications, reflect the use of public grant funding and extensive stakeholder engagement for smart cities, transit financing, and infrastructure innovation.

From the outset, Vivek distinguished himself not only through his technical ability but also through his initiative and capacity to navigate complex, interdisciplinary problems. Over the course of more than a year, we worked closely on projects focused on smart cities, freight mobility, and the validation of infrastructure data using advanced methods, including blockchain and artificial intelligence.

Vivek's impact on our work has been substantial. He played a central role in automating complex analyses for California's innovative freight initiatives, developing Python-based simulation models that significantly enhanced the efficiency and scalability of our research. He brought strong data-engineering and analytical skills to our team, allowing us to address emerging transportation

challenges more effectively. His work spanned transportation data validation, economic optimization of rail infrastructure, and accessibility modeling for low-income transit users.

In particular, Vivek applied advanced Python-driven simulation and optimization techniques to grade separation projects, using bi-objective and multi-objective frameworks to balance cost, safety, and operational performance. His methods combined technical rigor (e.g., regression analysis, Pareto-front modeling) with a holistic view of economic and operational impacts. This led to innovative solutions that have been referenced by public agencies and in the academic literature. His research has been cited in multiple international contexts, including studies of rail investment and transport equity in regions such as India, China, Australia, and Europe, as published in journals like *Sustainability* and the *Open Transportation Journal*. His contributions informed grade separation policies, value-capture funding models, and accessibility improvement strategies for agencies such as Caltrans, the California Public Utilities Commission (CPUC), and the Federal Transit Administration. Key outcomes included modeling benefit–cost ratios for infrastructure investments, supporting policy frameworks for Enhanced Infrastructure Financing Districts, and developing practical tools to assess and mitigate transport inequality.

Another defining strength is Vivek’s ability to operate effectively at both technical and strategic levels. He routinely led project workstreams independently, shaped the direction and scope of our deliverables, and proposed clear, actionable solutions under tight timelines—traits characteristic of someone at the forefront of his discipline. He oversaw smart city integration analyses, multimodal transportation assessments, and economic benefit evaluations. His responsibilities included designing and implementing simulation models, developing new analytical methodologies (such as an inequality-to-investment ratio), and communicating results to multi-agency stakeholders in a clear and compelling way. His leadership extended to mentoring junior team members, synthesizing interdisciplinary findings, and coordinating deliverables to ensure that our work translated into real-world impact on national and local infrastructure initiatives.

What truly sets Vivek apart—and why I believe his work directly advances U.S. national interests—is his rare combination of technical depth and operational insight. His experience integrating fragmented systems, automating infrastructure monitoring, and building multidisciplinary capabilities aligns closely with the challenges currently facing American utilities and critical infrastructure operators. These are not isolated research achievements; they demonstrate a sustained ability to bridge the gap between advanced technology and practical systems planning, thereby supporting resilient, future-ready public services.

The United States is investing heavily in AI-enabled technologies and smart infrastructure, with smart cities, advanced mobility, and resilient supply chains occupying a central place in policy and public discourse. In this context, Vivek’s track record of solving problems that mirror the most pressing challenges in American infrastructure is particularly valuable. He possesses the skills needed to integrate complex systems, future-proof operations, and enhance the reliability of essential services. His work consistently bridges technical excellence and operational practicality, generating long-term value and resilience for both public agencies and private-sector partners.

Vivek’s expertise is therefore uniquely aligned with the current priorities and direction of the United States. There is clear national momentum around advancing smart cities, freight mobility,

and infrastructure validation through technologies such as blockchain, AI, and real-time data analytics. Federal agencies, including the Department of Transportation and the Federal Highway Administration, are expanding their leadership in these areas, while cities and states nationwide are undertaking projects to improve efficiency, security, and resilience. Vivek's background, skills, and demonstrated record position him to contribute immediately and meaningfully to these urgent and ambitious initiatives.

Having observed his work firsthand and seen its direct impact on our research program and the broader field, I would welcome any opportunity to collaborate with Vivek again. His professionalism, integrity, and problem-solving ability are exceptional and rare. The United States would benefit greatly from his presence and continued contributions.

I recommend Vivek Mishra without reservation. He is precisely the kind of leader, innovator, and technical expert who can help shape the future of American infrastructure for the better. Our ongoing collaboration is well documented through joint publications, funded projects, and policy work on rail transport equity, grade separation, and sustainable urban mobility. Looking ahead, we anticipate focusing even more on Intelligent Transportation Systems. For upcoming major rail projects in California—including high-speed rail and key intercity corridors—Mr. Mishra's expertise in AI, real-time data analytics, predictive modeling, and safety systems will be critical for delivering resilient, equitable, and sustainable outcomes.

Please feel free to contact me at [shailesh.chandra@csulb.edu](mailto:shailesh.chandra@csulb.edu) if you have any questions or require further information.

Sincerely,



**Shailesh Chandra, Ph.D.**

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